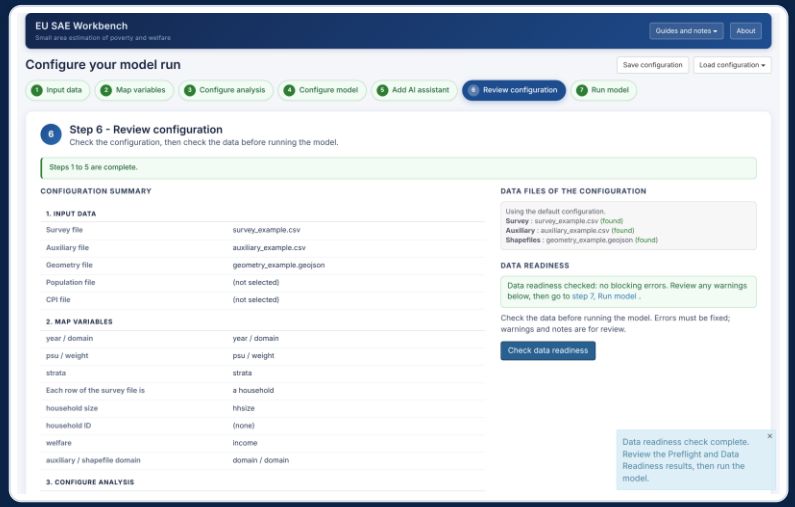
EU SAE WORKBENCH · GUIDE 2 OF 3

Running the Workbench

The seven guided steps, from your data files to the report.



OVERVIEW

Seven steps from data to results

Each step is one screen. You can move freely between them; Run model asks for a data readiness check first.

1. **Input data**
   Select the survey, auxiliary and map files
2. **Map variables**
   Say which column holds which variable
3. **Configure analysis**
   Two years and the indicator
4. **Configure model**
   UFH, MFH and benchmarking
5. **Add AI assistant**
   Optional report text
6. **Review configuration**
   Summary and data readiness check
7. **Run model**
   Progress, then the report

Front page: New model run starts at Step 1 with default settings. Load saved configuration fills every step and opens Step 6.

The classic dashboard (Start_Dashboard) has the same settings on one page and runs the same analysis.

Three files are required

They can be in any folder. Use the same domain codes in all three.

Survey

Household or person records

- year, domain code, PSU, survey weight and welfare
- household size, when each row is a household
- welfare per person (per capita or equivalised), not the household total
- optional: strata, household ID, poverty line, benchmark group

Auxiliary

Area covariates

- one row per domain and year
- domain code, year and candidate covariates
- covers both analysis years
- may include areas without survey data

Geometry

Area boundaries

- one polygon per domain
- a domain code matching the other two files
- shapefile: one .zip with the .shp, .shx and .dbf

Formats: R (.rds, .RData), CSV and other text files, Stata, SPSS, SAS, Parquet/Feather and Excel. Geometry: .rds, GeoPackage, GeoJSON, KML/GML or a zipped shapefile.

Optional files

Add them only when the analysis needs them.

Consumer price index (Step 1)

For mean welfare in constant prices. One row per year: a year column and the index. The values can also be typed in at Step 3.

Domain population sizes (Step 1)

For benchmarking: one row per domain-year (long) or domain (wide). Duplicate keys are rejected, even for equal totals. Without a file, use survey totals.

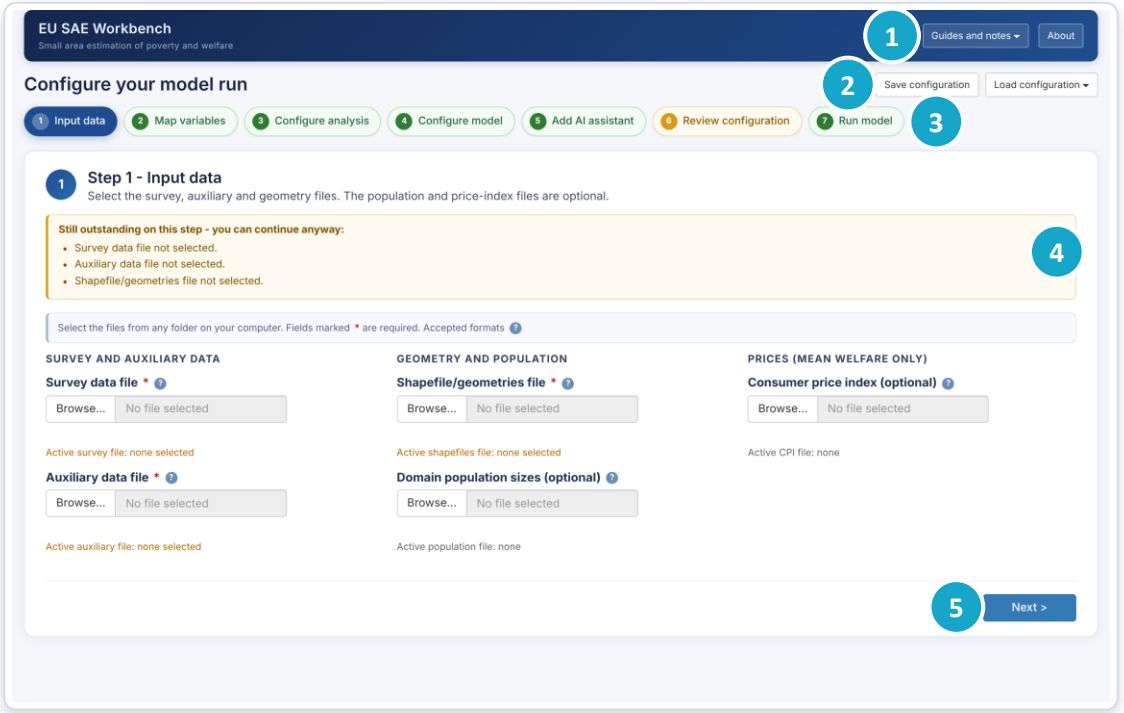
Benchmark targets (Step 4)

External targets: one row per group-year (long) or group (wide). Duplicate keys are rejected, even for equal values. Without a file, use survey Direct estimates.

Named matrix columns match years. Unnamed columns require exactly the selected years, in order, with a warning.

The screen at a glance

Every step has the same layout. Step 1 is shown before any file is selected.



1. **Guides and notes, About**
   Open the guides from inside the Workbench.

2. **Save and load configuration**
   Keep all settings for later or for another run.

3. **Step buttons**
   Green: done. Amber: missing items. Blue: current. Click to jump.

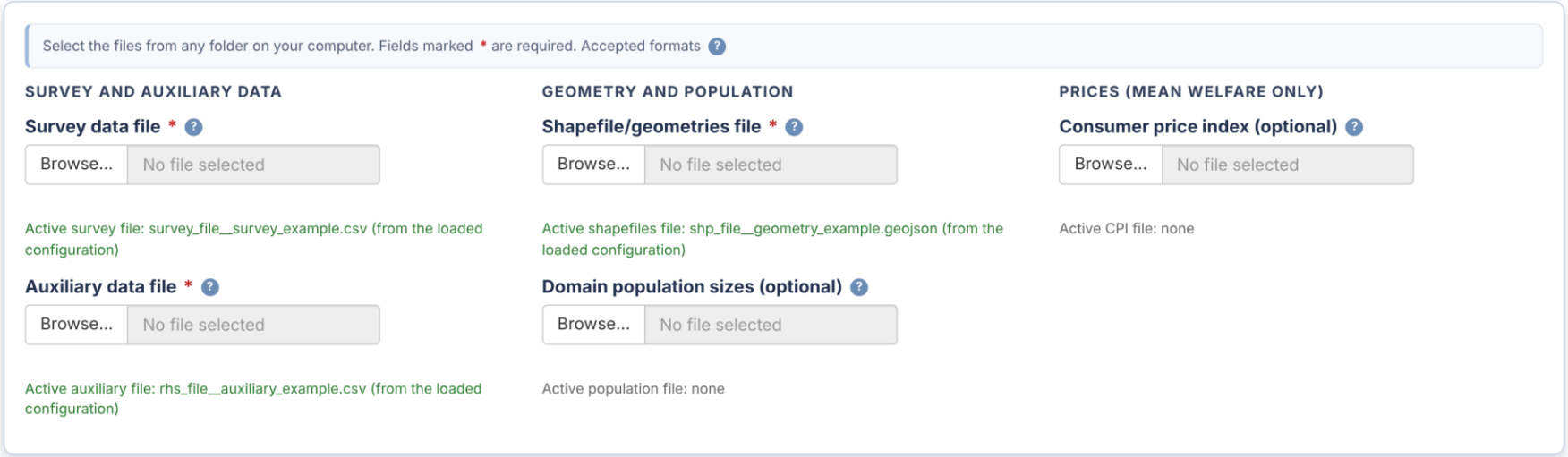
4. **What is still missing**
   Listed on each step. You can continue anyway.

5. **Back and Next**
   Move through the steps in order.

STEP 1

Input data

Browse to each file. Required fields carry a red asterisk.



Required
Survey, auxiliary and geometry files.

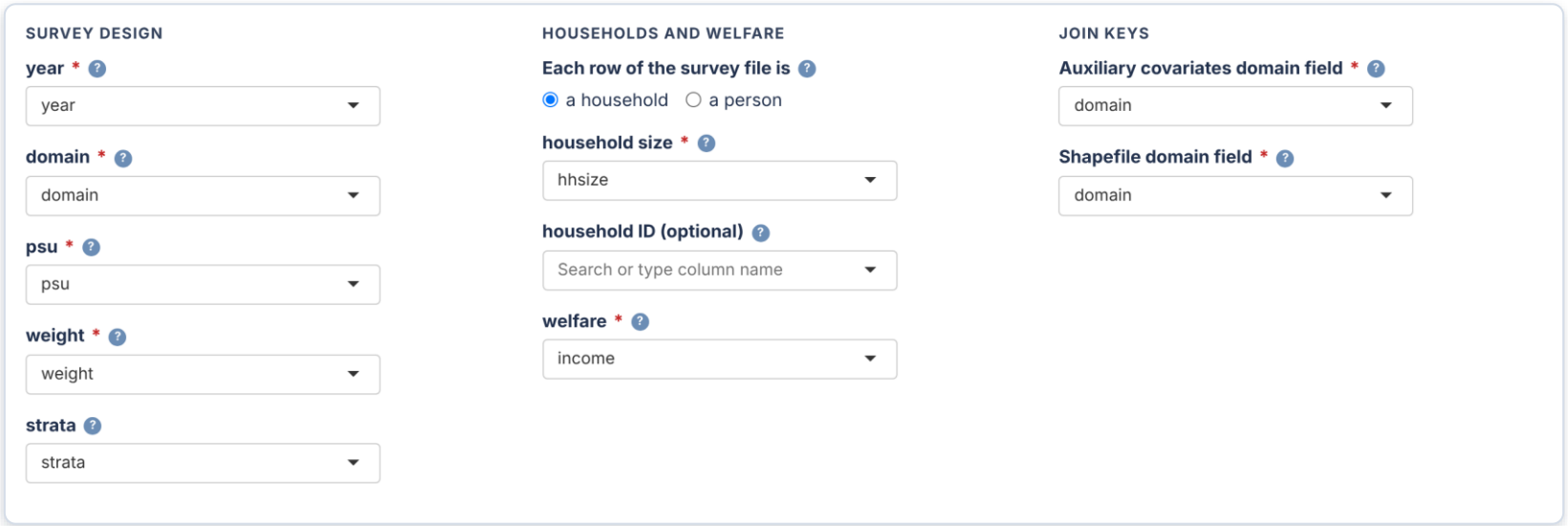
Optional
Domain population sizes; a consumer price index for mean welfare.

After loading a configuration
Its files appear under each Browse button. Browse only to change one.

STEP 2

Map variables

Choose the column that holds each variable. The lists show the columns of your files.



Survey design
Year, domain, PSU and weight; strata if the design has them.

Households and welfare
What one row is, household size, and welfare per person.

A field in red
Names a column that is not in the selected file.

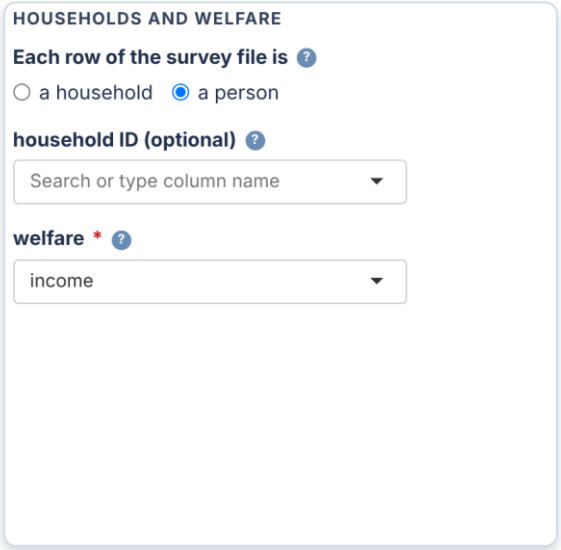
STEP 2

Households or persons?

Tell the Workbench what one row of the survey file is. This decides how people are counted.



A household: household size is required



A person: household size is not used

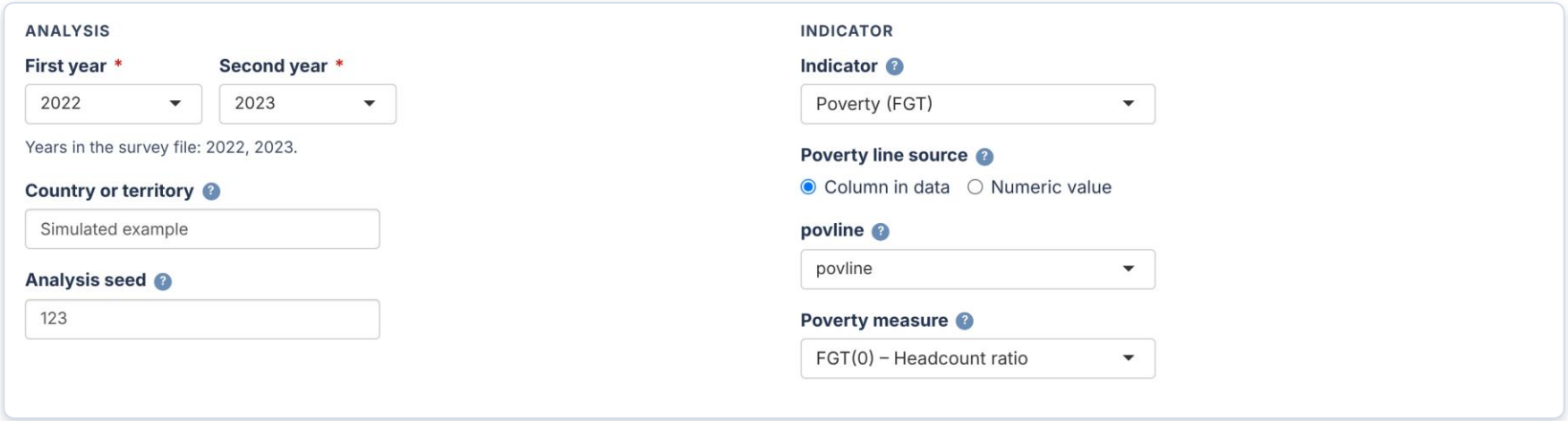
- **A household:** weight × household size counts every member.
- **A person:** the weight is used as it is.
- **Household ID (optional):** lets the readiness check confirm the choice.

A person file read as households counts each person once per household member.

STEP 3

Configure analysis

Pick the two years to compare, then the indicator.



Two years
Chosen from the years found in the survey file.

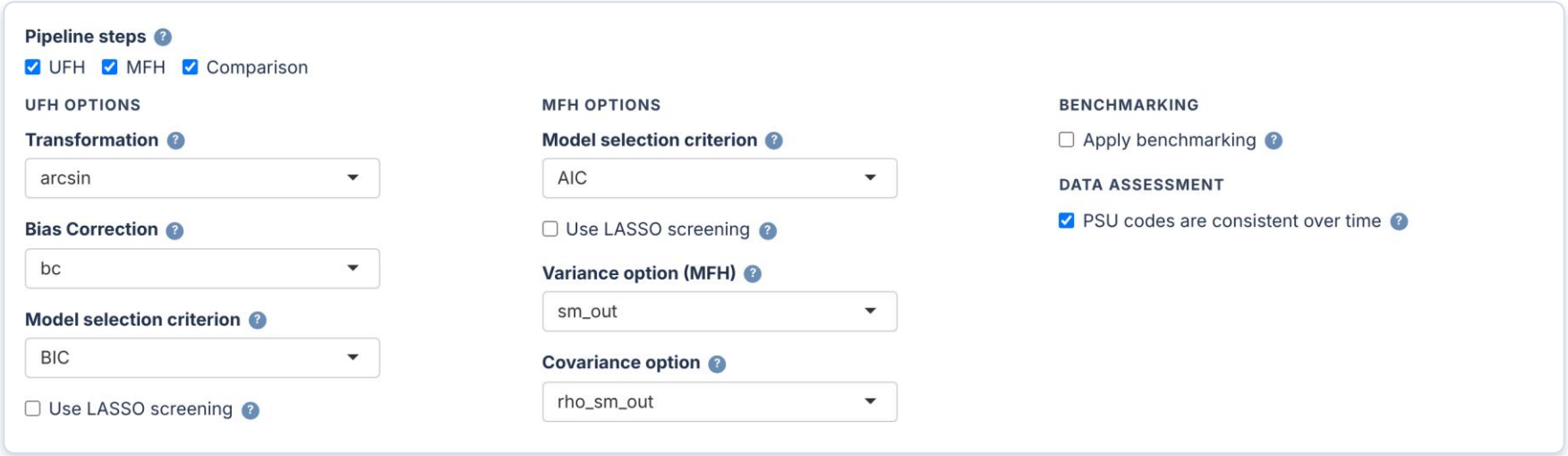
Analysis seed
The same seed, data, settings and package versions give the same results.

Indicator
A poverty measure (FGT) or mean welfare. Guide 3 covers both.

STEP 4

Configure model

Current defaults below. The screenshot retains historical w5o settings.



Pipeline steps
UFH: one model per year. MFH: both years jointly. Comparison: both side by side.

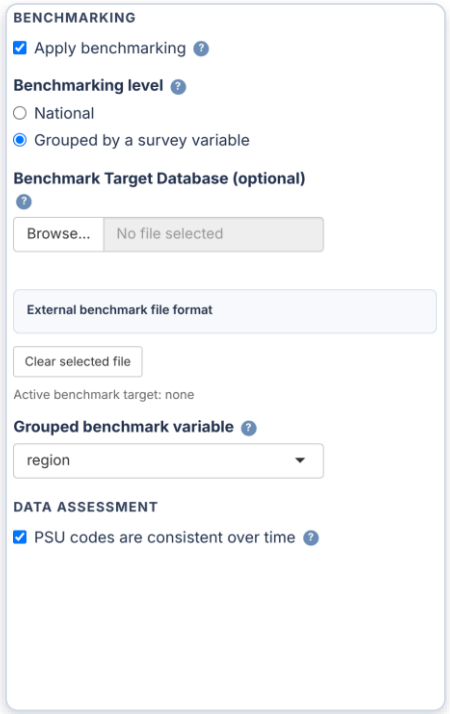
Defaults
MFH2, sm_out / rho_sm_out, 500 replicates. Explicit saved choices are retained.

Cross-year PSU design
Choose from the survey design. MFH does not run while it is Not specified.

STEP 4

Benchmarking (optional)

Makes the population-weighted average of the domain estimates equal the national or group figure.



- **National:** one constraint for the whole country.
- **Grouped by a survey variable:** one constraint per group, for example region or NUTS2.
- **Each domain must belong to one group.** If a domain has two group codes, the readiness check names it, its codes, years and rows.
- **Target file (optional):** same indicator and prices as the survey. Without it, the survey's direct estimates are the targets.

STEP 5

Add AI assistant (optional)

Adds plain-language interpretation and an advisory UFH-or-MFH recommendation. Off by default.



Consent first
Tick Enable and the consent box, then paste a Claude (sk-ant-…) or OpenAI (sk-…) key.

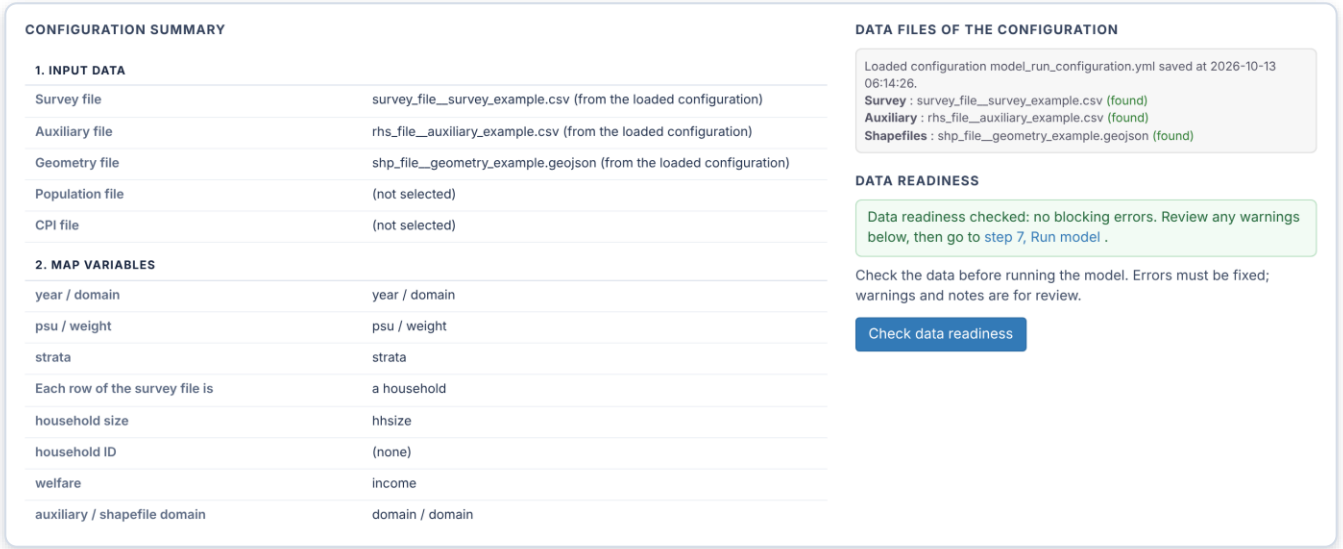
What is sent
Aggregate results only: no microdata, no area names or codes.

Key, language, Comparison
The key is not saved. 24 EU languages and Arabic. Runs with the Comparison step.

STEP 6

Review configuration

Read the summary, then press Check data readiness.



CONFIGURATION SUMMARY

1. INPUT DATA	
Survey file	survey_file__survey_example.csv (from the loaded configuration)
Auxiliary file	rhs_file__auxiliary_example.csv (from the loaded configuration)
Geometry file	shp_file__geometry_example.geojson (from the loaded configuration)
Population file	(not selected)
CPI file	(not selected)
2. MAP VARIABLES	
year / domain	year / domain
psu / weight	psu / weight
strata	strata
Each row of the survey file is	a household
household size	hhsize
household ID	(none)
welfare	income
auxiliary / shapefile domain	domain / domain

DATA FILES OF THE CONFIGURATION

Loaded configuration model_run_configuration.yml saved at 2026-10-13 06:14:26.
Survey : survey_file__survey_example.csv (found)
Auxiliary : rhs_file__auxiliary_example.csv (found)
Shapefiles : shp_file__geometry_example.geojson (found)

DATA READINESS

Data readiness checked: no blocking errors. Review any warnings below, then go to step 7, Run model .

Check the data before running the model. Errors must be fixed; warnings and notes are for review.

Check data readiness

Summary
Every setting on one page, and where each data file comes from.

Check data readiness
Reads the data and runs the checks. The results appear below the summary.

Changed something?
Check again: the app asks for it after changes to data, mapping, years, indicator or prices.

STEP 6

Reading the data readiness results

ERROR lines must be fixed before the run. WARNING lines and notes are for review.

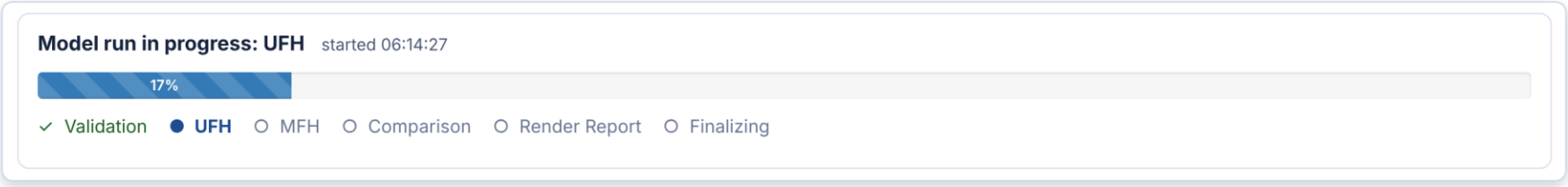
Test	What it checks
0a, 0b	A year column in both files; both analysis years present in the survey and auxiliary files
0c	Weights and household sizes give valid population weights
0d, 0g	Grouped benchmarking: the group variable exists; each domain belongs to one group
0e	Mean welfare in logs: zero or negative welfare, and positive area means
0f	Households or persons: the weighted population total and, with a household ID, the row type
0h	Consumed population/target inputs: unique keys, domain/group coverage and selected years
1	Covariates: means and correlation with the indicator
2, 3	Domain coverage: MFH requires a Direct estimate for every domain in both years
4	The national poverty rate or mean welfare in each year

MFH missing domain-years and invalid required inputs block the run before output cleanup. Unused optional paths do not block it.

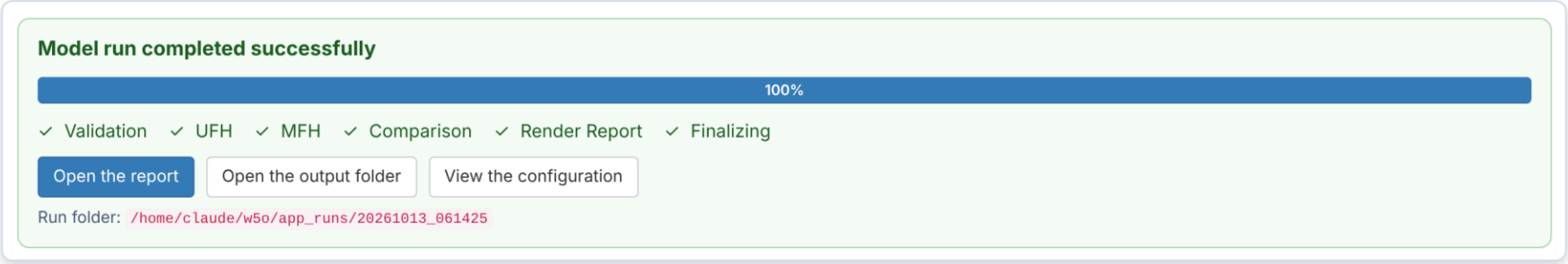
STEP 7

Run model

The run continues in the background. The page shows each stage, then the result.



While the model runs: the stage and its progress



When it is complete: open the report, the output folder or the configuration

Stop ends a run. If a run fails, the panel shows the reason and a button to open the run folder.

RESULTS

Where the results are

outputs/ holds the latest run. Every run is also kept in its own folder.

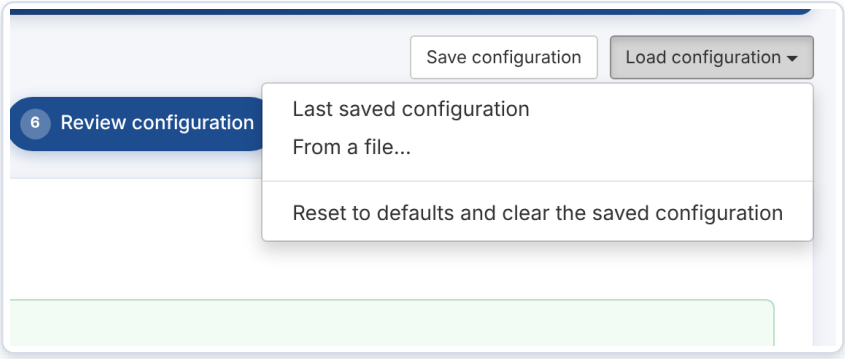
File or folder	What it contains
outputs/final_report.html, .docx	The report: maps, tables, diagnostics and changes. The Word file is editable.
outputs/data/	Excel workbooks with Direct, UFH and MFH estimates, MSE, RMSE and CV, and the changes
outputs/figures/	Maps and charts as PNG files
outputs/tables/	CSV tables: readiness, fit checks, bootstrap diagnostics and ufh_variance_audit.csv
app_runs/<date_time>_<label>/	A copy of the run's outputs, its log and model_run_configuration.yml

A validated run replaces outputs/. To edit the Word report, work on a copy, or use the copy under app_runs/.

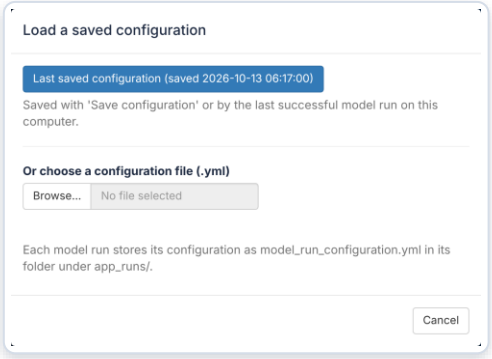
Direct MSE is raw survey-design variance. ufh_variance_audit.csv separately records the variance used by the model, its scale and policy.

Save, load and repeat a run

A configuration holds every setting and where the data files are. Reuse it instead of starting again.



Load configuration menu, on every step

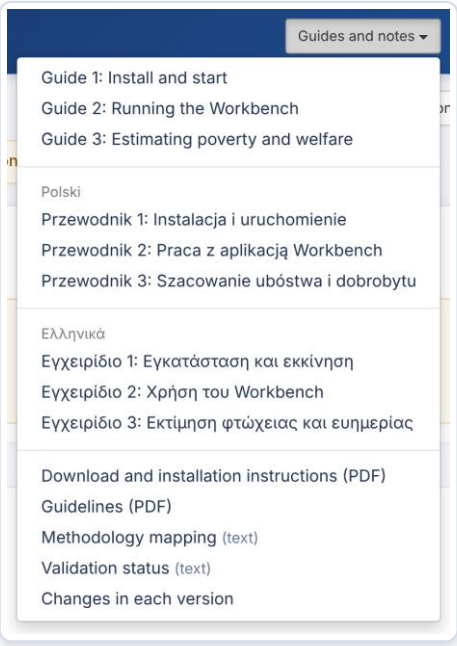


Load saved configuration, on the front page

- **Save configuration:** keeps the settings and copies of the data files in app_runs/_last_setup_files.
- **Load:** the last saved configuration, or a file such as a run's model_run_configuration.yml. The review step opens.
- **Changed data file?** Loading warns if a file differs from the one the configuration was saved with.
- **Reset to defaults** deletes the saved file copies; older run configurations then need their files browsed again.

Help inside the Workbench

Three places to look before asking.



Guides and notes menu

? **Question marks**
Hover over the ? next to a field for what it means and which values it accepts.

G **Guides and notes**
The three guides in English, Polish and Greek, the guidelines note, the methodology mapping, the validation status and the changes in each version.

i **About**
The version you are using.

Problem reports: github.com/Noboyoshida227/eu-sae-application/issues. Attach the run log, never the data.

CHECKLIST

Every run, in short

Before you report results, check more than the map.

Step	Check
1 Input data	The three required files; CPI or population file if needed
2 Map variables	No red fields; household or person rows; household size
3 Configure analysis	Two years from the survey; indicator and poverty line or prices
4 Configure model	Defaults unless you have a reason; PSU answer; benchmarking
5 Add AI assistant	Off, or consent and key
6 Review configuration	Summary read; readiness without ERROR
7 Run model	Report opened; diagnostics, bootstrap results and changes reviewed

Guide 3 shows the settings and the reading of results for poverty, anchored poverty and mean welfare.